

Deepfake Trust & Remote Work

Why Seeing and Hearing No Longer Means Verifying

Cyber Psychology Portfolio Project
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Research Question: How do people calibrate trust in identity and authenticity during remote interactions, and how does synthetic media exploit that trust?

1. Executive Summary

For most of modern working life, seeing a colleague's face on a video call or recognizing a supervisor's voice on the phone has served as reliable proof of identity, not because anyone reasoned it out, but because faking a live face and voice was conventionally difficult and expensive. Generative AI has quietly removed that constraint. This report examines the psychological mechanisms that lead people to trust audio and video as verification, and how deepfakes and voice-cloning technology exploit those mechanisms in a remote-work setting.

The core findings are that this is not primarily a technology-literacy problem. People are not failing to trust video calls because they don't understand how strong the effects exist. They are trusting video calls because trusting familiar faces and voices is a deep, automatic, largely unconscious process that evolved for a world where it was safe to extend trust. The fix, correspondingly, is not more skepticism in general but a specific, procedural habit: treating identity confirmation only through audio or video, however convincing. Yes, it's efficient for high-stakes requests without an independent, out-of-band check.

2. Background

In January 2024, a finance employee at the multinational engineering firm Arup was drawn into a voice conference call with people who appeared and sounded like the company's UK-based CFO and several colleagues. Every participant on the call was an AI-generated deep-fake, built from publicly available video and audio of real executives. Believing the call was genuine, the employee carried out 15 transfers totaling roughly \$25.6 million USD to accounts controlled by the attackers. The scheme began. Notably, while the ordinary phishing email was sent, the deepfake video call was what overcame the employees' initial skepticism about the email.

The case illustrates the central shift this report addresses: an attack is no longer needed to impersonate authority through text alone. They can now impersonate the specific sensory cues of a recognized face or voice that people have always reasonably relied on as strong evidence of identity.

3. Psychology Framework

3.1 Truth-Default Theory

Communication researcher Timothy Levine's truth-default theory holds that people operate from a default assumption that others are being honest, and that this default is adaptive. Constantly scrutinizing everyone for deception would make ordinary cooperation and communication impossible. The default is only overwritten by a specific trigger that raises suspicion. A convincing deep fake is specifically engineered to avoid triggering that override.

3.2 The Modality Trust Heuristic

People implicitly rank Communication channels by how hard they are to fake: text is easiest to fake, a phone call is harder, and a live video call with a recognizable face is the hardest of all. This ranking was accurate for most of communication history, so it became an unconscious heuristic that more "real-seeming" modalities receive proportionally less scrutiny. Generative AI has broken the link between how a message arrives and how hard it is to fake, but the heuristic has not caught up.

3.3 Authority and Urgency, Amplified

The same triggers documented elsewhere in this portfolio, authority and urgency, are present in deep fake fraud. Still, the synthetic video or voice acts as a force multiplier: it supplies visuals and auditory "proof" of the authority's claim that a text-only message never could. In the Arup case, seeing and hearing colleagues on the call resolves the employee's doubt about the original fishing email rather than raising suspicion.

3.4 Divided Attention in Remote Work

Video calls routinely compete with chat notifications, second monitors, and multitasking, leaving me less cognitive bandwidth for the kind of close, detailed observation that might catch subtle deepfake artifacts, unnatural blinking, lip-sync drift, or inconsistent lighting. Remote work, under normal conditions, is, incidentally, conducive to defects going unnoticed.

3.5 Overconfidence in Voice and Face Recognition

People are systematically confident in their ability to recognize familiar voices and faces, especially under time pressure or in professional contexts where questioning a superior's identity feels socially costly. That overconfidence is precisely what allows a merely good deepfake, rather than a perfect one, to succeed.

4. Fictional Application: Elite Healthcare

Applying this framework to Elite Healthcare, the fictional organization used elsewhere in this portfolio, illustrates how the same attack pattern will unfold outside of the finance sector.

Scenario

A building department employee receives a message that appears to come from at least a healthcare CFO, requesting an urgent, confidential change to a vendor's payment account ahead of a system migration deadline. When the employee hesitates, a follow-up video call is proposed. The "CFO" joins on camera, along with a "colleague" from procurement, the employee recognized from prior meetings. Both are AI-generated. The call is brief, framed as a formality to reassure the employees, and ends with the CFO reiterating the urgency and stepping away for "another meeting," Leaving the procurement "colleague" to finalize details over chat.

5. Analysis of Manipulation Techniques

Stage	Technique	Psychological Principle
Initial message	Authority claim + confidentiality framing	Truth- default theory; confidentiality discourages employees from casually verifying with others
Employee hesitates	Video call offer proactively, as reassurance	The modality trust heuristic video is offered specifically because it resolves doubt more effectively than more text.
During the call	Recognizable face in voice, familiar colleague present	Overconfident in face/voice recognition; social cause of questioning a superior on camera
Call framed as brief	CFO leaves quickly, limiting observation time	Divided attention / limited window to notice subtle inconsistencies
Hand off to chat	Urgency reiterated, details finalized off-call	Urgency: Moves the actual instructions back to the easiest-to-fake channel, once trust is established

6. Recommendations

6.1 Individual-Level

- Treat a video or voice call as identity confirmation for routine matters, but not as sufficient authorization for financial, credential, or data-access changes on its own, still requires independent verification.
- If a request is confidential, urgent, and financial at the same time, treat that specific combination as a cue to slow down, regardless of who appears to be asking.
- It is not socially inappropriate to verify a superior identity through a separate channel; frame this internally as standard procedure rather than a personal accusation.

6.2 Organizational-Level

- Require independent, out-of-band verification (a call back to a known number, not one provided in the request itself) for any payment, account, or credential change, regardless of how the request was confirmed on video or by phone.
- Establish a share verification phrase or challenge-response method for high-authority individuals to use in any request involving money or sensitive access, to be reviewed and rotated periodically.
- Build a formal, blame-free process to verify an executive's identity, so employees are not forced to choose between following instructions and risking an awkward confrontation.
- Extend awareness training to cover synthetic media, specifically not just fishing, so staff understand that a familiar face or voice is no longer sufficient proof on its own.
- Limit how much of high-authority individuals' voice and video is publicly available (recordings of town halls, conference talks, social posts). These are the raw materials from which defects are built.

7. Conclusion

Deepfake-enabled fraud does not succeed by defeating people's judgment. It succeeds by using the exact evidence people have always, reasonably, relied on to make that judgment. The response is not to make employees more suspicious of everyone by default, which is neither realistic nor healthy, but to move high-stakes verification off the channel being impersonated and onto a separate, procedural check that does not depend on trusting a face or voice in the moment. That shift from "I saw and heard them" to "I confirmed it independently" is the throughline connecting this project to the rest of the portfolio.

